

Monopile Installation Record HB-A04

Harrow Bay Offshore Wind · Foundations

Elena Márquez · 27 April 2026 · HB-CON-IR-A04

Position	Date	Vessel	Contract
HB-A04	25 Apr 2026	MV Sea Atlas	WE-CON-2025-040 / -027

1. Position and pile data

Item	Design	As-installed
Coordinates (WGS84)	53.91830 N, 1.08140 E	within 23 cm of target
Water depth (m LAT)	30.3	30.4
Pile length (m)	72.8	72.8
Pile mass (t)	1,257	–
Target penetration (m)	35.1	35.2
Verticality	≤ 0.25°	0.17°

2. Driving record

Item	Value
Start of soft start	12:06
Driving duration	4 h 26 min
Total blow count	5,390
Maximum hammer energy	2,899 kJ
Self-weight penetration	3.8 m
Interruptions	1 × 25 min (hammer hose check)

3. Noise and marine mammal mitigation

Item	Value
Bubble curtain	Double big bubble curtain, both rings operational
Peak SEL at 750 m	160.6 dB (limit 160 dB)
Pre-piling search (MMO + PAM)	30 min, clear
Acoustic deterrent device	Activated 30 min before soft start

Noise limit exceeded (160.6 dB). Regulator notified within 24 h. Corrective action: hammer energy capped at 3,000 kJ, hose re-laid on tighter radius (decision D-008).

4. Transition piece

Item	Value
TP installed	27 Apr 2026
Flange flatness after installation	1.4 mm (limit 2.0 mm)
Bolt tensioning	148 × M72 – completed
TP level	+0.03° (limit 0.25°)

5. Non-conformances

NCR-HB-086: minor coating damage on pile top during upending, repaired offshore.

6. Sign-off

Role	Name	Date
Offshore Installation Supervisor	Atlas Marine Contractors	26 Apr 2026
Client Representative	Felix Brandt	26 Apr 2026
Reviewed	Elena Márquez	27 Apr 2026